

Problem

Find $v(t)$ in the following integrodifferential equations using the phasor approach:

(a) $v(t) + \int v \, dt = 10 \cos t$

(b) $\frac{dv}{dt} + 5v(t) + 4 \int v \, dt = 20 \sin(4t + 10^\circ)$

Step-by-step solution

Step 1 of 3

(a)

Consider the integrodifferential equation.

$$v(t) + \int v \, dt = 10 \cos t$$

Convert the time domain expression into the frequency domain expression.

$$\mathbf{V} + \frac{\mathbf{V}}{j\omega} = 10 \angle 0^\circ$$

Substitute 1 rad/s for ω .

$$\mathbf{V} + \frac{\mathbf{V}}{j(1)} = 10$$

$$\mathbf{V}(1 - j) = 10$$

Step 2 of 3

Simplify further.

$$\begin{aligned} \mathbf{V} &= \frac{10}{1 - j} \\ &= \frac{10}{\sqrt{2} \angle -45^\circ} \\ &= \frac{10}{\sqrt{2}} \angle 45^\circ \end{aligned}$$

Convert the phasor expression into time domain.

$$\begin{aligned} v(t) &= \frac{10}{\sqrt{2}} \cos(t + 45^\circ) \\ &= 7.071 \cos(t + 45^\circ) \end{aligned}$$

Thus, the expression of $v(t)$ is $\boxed{7.071 \cos(t + 45^\circ)}$.

Step 3 of 3

(b)

Consider the integrodifferential equation.

$$\frac{dv}{dt} + 5v(t) + 4 \int v dt = 20 \sin(4t + 10^\circ)$$

$$\frac{dv}{dt} + 5v(t) + 4 \int v dt = 20 \cos(4t + 10^\circ - 90^\circ)$$

Convert the time domain expression into the frequency domain expression.

$$j\omega \mathbf{V} + 5\mathbf{V} + 4 \frac{\mathbf{V}}{j\omega} = 20 \angle -80^\circ$$

Substitute 4 rad/s for ω .

$$j4\mathbf{V} + 5\mathbf{V} + 4 \frac{\mathbf{V}}{j4} = 20 \angle -80^\circ$$

$$\mathbf{V}(j4 + 5 - j) = 20 \angle -80^\circ$$

$$\mathbf{V} = \frac{20 \angle -80^\circ}{5 + j3}$$

Simplify further.

$$\begin{aligned} \mathbf{V} &= \frac{20 \angle -80^\circ}{\sqrt{34} \angle 30.96^\circ} \\ &= 3.43 \angle -110.96^\circ \end{aligned}$$

Convert the phasor expression into time domain.

$$v(t) = 3.43 \cos(4t - 110.964^\circ)$$

Thus, the expression of $v(t)$ is $\boxed{3.43 \cos(4t - 110.96^\circ)}$.